ZULHELMI SYAHTIAR

Fresh Graduate D3 Industrial Electronics

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SUMMARY

Fresh graduate in Industrial Electronics from Politeknik Negeri Jakarta with internship experience in power system protection at PT PLN. Strongly interested in the Internet of Things (IoT), microcontrollers, programming, and control systems. Skilled in hands-on microcontroller-based projects and simple automation system development. Eager to continuously learn, innovate, and apply modern technologies in industrial environments.

EXPERIENCE

PT PLN (Persero) ULTG Petukangan – Tangerang Selatan *Jul 2025 – Nov 2025*
Protection Maintenance Intern (HAR PRO)

- Assisted with periodic maintenance and monitoring of substation protection systems.
- Performed testing of protection relays (OCR, GFR, Differential, Distance Relay, OVR/UVR, Buchholz, Thermal Overload).
- Involved in system testing during DSS migration at a 150kV substation.
- Carried out Battery Capacity Tests (BCT) to maintain the reliability of substation backup power.
- Assisted with non-cascade testing on transformer incoming panels.

Independent Clients – Jabodetabek *Mar 2023 – Present*
Freelance Graphic Designer & Photographer

- Provided graduation and post-thesis-defense photography services, including style direction and photo editing.
- Created banner designs for graduation celebration events.
- Created social media graphic designs for organizational Instagram content.

EDUCATION

Politeknik Negeri Jakarta – Depok, Jawa Barat *2023 – 2026*
D3 Industrial Electronics – GPA 3.55/4.00

- Final Project: "Dual DC Motor Control Trainer Kit – PID – Fuzzy Control Based on ESP32 with 3D Digital Twin Visualization".
- Teaching Assistant for a Community Service project installing a solar-powered Hydroponic Green House.
- Board Member of KSM PSYCHOROBOTIC, Communication & Information Department (2025).
- Committee member, Publication, Documentation & Creative Media Division at E-TIME 2025, Olahraga Elektro 2024, FRACTION 2024 (Electrical Engineering Department Graduation), and the Industrial Electronics Study Excursion.
- Committee member, Media Division at the EC-IKI PNJ 2024 Public Lecture.

SMAN 64 Jakarta – Jakarta Timur *2019 – 2022*
High School – Science Major (MIPA)

CERTIFICATIONS

- BNSP Competency Certification – Instrumentation Technician (2026) – Competent. Competency test: zero-span configuration and scaling of a level transmitter (4–20 mA) on a PLC, including linearity and measurement-deviation analysis.
- TOEIC Prediction Test, Politeknik Negeri Jakarta (2026) – Score 620/990 (Listening 260, Reading 360).

FEATURED PROJECTS

- DC Motor Control Trainer Kit (2025–2026): Web monitoring dashboard (JavaScript ES6, HTML5, Tailwind CSS v4) and an interactive 3D Digital Twin (Three.js r160) synchronized in real time with ESP32 hardware via Firebase Realtime Database.
- IoT Web Monitoring Dashboard (2025): Monitoring of temperature and soil & air humidity based on a MySQL database.
- Plant Monitoring Robot (2025): ESP32-C3 with DHT11 and soil moisture sensors, IoT-based.
- Logic Gate Training Module (2024): Practice kit for AND, OR, NOT, NAND, NOR, XOR, Full Adder, Multiplexer, Demultiplexer, Flip-flop, and BCD-to-7-Segment Decoder.
- 9V Power Supply based on IC 7809 (2024): Design, assembly, and testing of a stable power supply for small-scale electronic devices.
- Object and Color Detection Mobile Robot (2026): Developed a real-time object and color detection system using a Raspberry Pi-based camera.
- Automatic Hand Washing Machine (2024): Designed an Arduino Uno and infrared sensor-based touchless system to automatically dispense soap/water.
- Numa Space (2026) A virtual study room and real-time study-buddy matchmaking platform. Enables P2P video/audio interaction via WebRTC, Socket.IO chat communication, a private room access code system, and Pomodoro Timer integration for productivity.

SKILLS

- IoT & Microcontrollers: ESP32, Arduino, Raspberry Pi, Firebase (Realtime Database, Firestore & Hosting), Computer Vision (OpenCV).
- Control & Automation Systems: PID, Fuzzy Logic Controller, PLC programming, SCADA (Schneider Electric, Vijeo Citect).
- Protection & Testing Systems: Protection Relays (OCR, Differential, REF MiCOM), OMICRON TestUniverse, GOOSE Configuration.
- Programming: Python, C/C++, HTML, CSS, JavaScript, TypeScript, React, Node.js, Express.
- Design & Simulation: NI Multisim, Proteus, AutoCAD, Fusion 360, Eagle.
- Visualisasi & Real-time: Three.js, WebRTC, Socket.IO.
- Database: MySQL, PostgreSQL.
- Design & Multimedia: Adobe Photoshop, Adobe Lightroom, Canva, CapCut
- Microsoft Office: Word, Excel, PowerPoint.
- Languages: Indonesian (native), Malay, English (TOEIC Prediction 620/990 – working proficiency).